computing within their existing IT infrastructure. The name 'Eucalyptus' stands for 'Elastic Utility Computing Architecture for Linking Your Programs to Useful Systems'.

What makes Eucalyptus stand out is its compatibility with Amazon Web Services (AWS). It provides a framework for building private and hybrid clouds that can mimic the behaviour of AWS services such as EC2 (Elastic Compute Cloud) and S3 (Simple Storage Service). This allows organisations to leverage AWS APIs and seamlessly integrate their on-premises infrastructure with public cloud resources provided by AWS.

As an infrastructure as a service (IaaS) solution, Eucalyptus allows organisations to deploy, manage, and scale computing resources such as virtual machines, storage, and networking services within a private cloud environment. It gives businesses the flexibility to retain control over their data and resources while also enabling them to take advantage of public cloud offerings when necessary. This makes Eucalyptus an ideal solution for organisations looking to create AWS-compatible private or hybrid cloud infrastructures without being locked into a single cloud provider.

History and evolution

Eucalyptus originated as a research project at the University of California, Santa Barbara in 2008. Its primary goal was to create an open source software platform capable of building private and hybrid cloud environments that could emulate the services provided by Amazon Web Services (AWS).

Eucalyptus became one of the first platforms to offer AWS API compatibility, making it highly attractive to organisations looking for a flexible, cloud-native solution while maintaining the option to integrate with AWS for hybrid cloud setups.

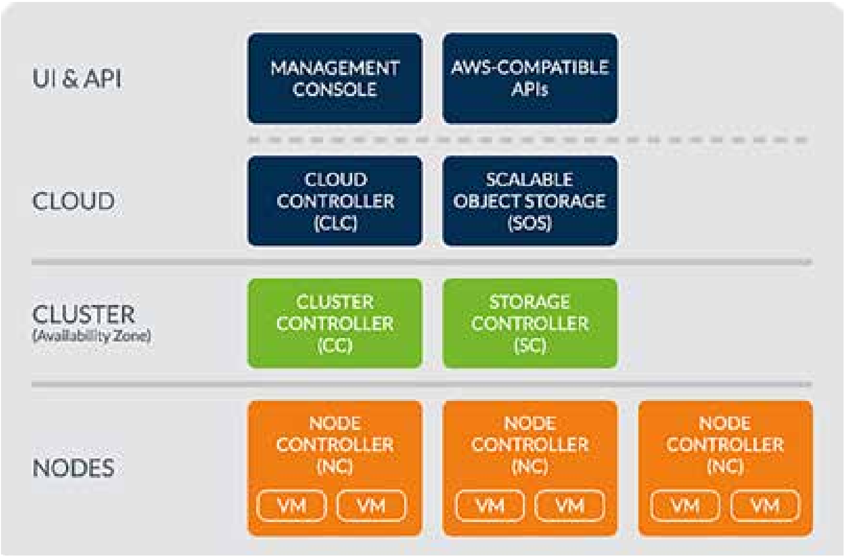

Figure 1: Eucalyptus cloud computing architecture

Initially designed for academic and research purposes, Eucalyptus rapidly gained attention from the industry due to its open source nature and its potential to bridge private infrastructure with public cloud services.

Here are the key milestones in Eucalyptus' evolution.

2009: Eucalyptus Systems, Inc. was founded to commercialise the platform and support its adoption in enterprise environments.

2012: Eucalyptus partnered with AWS, ensuring continued API compatibility between the two platforms. This was a significant step in reinforcing its position as a hybrid cloud solution.

2014: Hewlett-Packard acquired Eucalyptus with a vision to integrate its cloud offerings, particularly in hybrid cloud deployments. Despite the acquisition, the development and support for Eucalyptus continued.

2015 onwards: While newer platforms such as OpenStack gained traction, Eucalyptus has retained relevance due to its specialised AWS integration and ease of use for hybrid cloud scenarios.

Eucalyptus has since become a powerful tool for enterprises looking to implement cost-effective, AWS-compatible private clouds. Although the open source landscape has expanded with alternatives like OpenStack, Eucalyptus maintains a loyal user base due to its streamlined interface and specific use cases in hybrid cloud computing.

Key features of Eucalyptus

AWS API compatibility: Full compatibility with AWS services like EC2, S3, EBS, and IAM enables seamless integration and workload movement between private clouds and AWS.

Scalability: Eucalyptus allows dynamic allocation of compute, storage, and network resources, scaling to meet changing demands.

Flexibility: It supports both Linux and Windows, and can be deployed on commodity hardware, making it cost-efficient and customisable.

Security: Built-in user authentication, role-based access control, and AWS IAM integration ensure secure access and resource management.

Virtualisation support: Supports multiple hypervisors like KVM and Xen, providing flexibility in virtualisation.